§3.1 理想绝缘介质中的波动方程及平面电磁波解

- 波动方程

  介质中的电磁性质方程为

  $$\vec{B}=\mu\vec{H} \qquad\qquad \vec{D}=\epsilon\vec{E}$$

  理想绝缘介质中μ,ϵ为常数,特别地,$\mu=\mu_0$和$\epsilon=\epsilon_0$是真空. 本节考虑自由电荷和传导电流为零的情况,这时理想绝缘介质中的麦克斯韦方程为

  $$\nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t} \qquad \nabla\times\vec{B}=\mu\epsilon\frac{\partial\vec{E}}{\partial t} \qquad \nabla\cdot\vec{E}=0 \qquad \nabla\cdot\vec{B}=0$$

  计算

  $$\nabla^2\vec{E} \;=\; -\nabla(\nabla\cdot\vec{E})+\nabla^2\vec{E}=-\nabla\times(\nabla\times\vec{E})=\nabla\times\frac{\partial\vec{B}}{\partial t}=\frac{\partial}{\partial t}\nabla\times\vec{B}=\mu\epsilon\frac{\partial^2}{\partial t^2}\vec{E}$$

  类似地

  $$\nabla^2\vec{B} \;=\; -\nabla(\nabla\cdot\vec{B})+\nabla^2\vec{B}=-\nabla\times(\nabla\times\vec{B})=-\mu\epsilon\nabla\times\frac{\partial\vec{E}}{\partial t}=-\mu\epsilon\frac{\partial}{\partial t}\nabla\times\vec{E}=\mu\epsilon\frac{\partial^2}{\partial t^2}\vec{B}$$

  定义

  $$v^2=\frac{1}{\mu\epsilon} \qquad \text{特别地} \qquad c^2=\frac{1}{\mu_0\epsilon_0}$$

  则

  $$\nabla^2\vec{E}-\frac{1}{v^2}\frac{\partial^2\vec{E}}{\partial t^2}=0 \qquad\qquad \nabla^2\vec{B}-\frac{1}{v^2}\frac{\partial^2\vec{B}}{\partial t^2}=0$$

- 平面波解

  上述方程有通解

  $$\vec{E}(\vec{r},t)=\vec{E}(t-\frac{\vec{n}\cdot\vec{r}}{v}) \qquad\qquad \vec{B}(\vec{r},t)=\vec{B}(t-\frac{\vec{n}\cdot\vec{r}}{v})$$

  其中$\vec{n}$为常矢量,$\vec{n}\cdot\vec{n}=1$作为检验,我们计算

  $$\nabla\vec{E}(t-\frac{\vec{n}\cdot\vec{r}}{v})=\nabla(t-\frac{\vec{n}\cdot\vec{r}}{v})\frac{\partial\vec{E}}{\partial t}=-\frac{\vec{n}}{v}\frac{\partial\vec{E}}{\partial t} \qquad \nabla\cdot\nabla\vec{E}=-\frac{\vec{n}}{v}\cdot\nabla\frac{\partial\vec{E}}{\partial t}=\frac{\vec{n}}{v}\cdot\nabla(t-\frac{\vec{n}\cdot\vec{r}}{v})\frac{\partial^2\vec{E}}{\partial t^2}=\frac{1}{v^2}\frac{\partial^2\vec{E}}{\partial t^2}$$

  这个解具有以下特点:

  – 对给定时刻,等值面为与$\vec{n}$垂直的平面: $\vec{r}_\perp\cdot\vec{n}=0 \qquad \vec{n}\cdot\vec{r}=\vec{n}\cdot(\vec{r}+\vec{r}_\perp)$

  $$t-\frac{\vec{n}\cdot\vec{r}}{v}=t-\frac{\vec{n}\cdot(\vec{r}+\vec{r}_\perp)}{v}$$

– 等值面以速度v沿$\vec{n}$方向运动: 考虑两个时刻$t_1 < t_2$, 等值面由$\vec{r}_1$移到$\vec{r}_2$,则

$$t_1 - \frac{\vec{n}\cdot\vec{r}_1}{v} = t_2 - \frac{\vec{n}\cdot\vec{r}_2}{v} \qquad \rightarrow \qquad v = \vec{n}\cdot\left(\frac{\vec{r}_2-\vec{r}_1}{t_2-t_1}\right)$$

因此,此解给出的是沿$\vec{n}$方向以速度v传播的平面电磁波.

$$\nabla\times\vec{E} = \nabla(t-\frac{\vec{r}\cdot\vec{n}}{v})\times\frac{\partial\vec{E}}{\partial t} = -\frac{\vec{n}}{v}\times\frac{\partial\vec{E}}{\partial t} = -\frac{\partial\vec{B}}{\partial t}$$
$$\nabla\times\vec{B} = \nabla(t-\frac{\vec{r}\cdot\vec{n}}{v})\times\frac{\partial\vec{B}}{\partial t} = -\frac{\vec{n}}{v}\times\frac{\partial\vec{B}}{\partial t} = \mu\nabla\times\vec{H} = \mu\frac{\partial\vec{D}}{\partial t} = \mu\epsilon\frac{\partial\vec{E}}{\partial t} = \frac{1}{v^2}\frac{\partial\vec{E}}{\partial t}$$

它们化为

$$\frac{\partial}{\partial t}(v\vec{B}-\vec{n}\times\vec{E}) = 0 \qquad \rightarrow \qquad v\vec{B} = \vec{n}\times\vec{E} + \text{时间无关项}$$
$$\frac{\partial}{\partial t}(\vec{n}\times v\vec{B}+\vec{E}) = 0 \qquad \rightarrow \qquad v\vec{E} = -\vec{n}\times v\vec{B} + \text{时间无关项} = -\vec{v}\times\vec{B} + \text{时间无关项}$$

又

$$\nabla\cdot\vec{E} = \nabla(t-\frac{\vec{r}\cdot\vec{n}}{v})\cdot\frac{\partial\vec{E}}{\partial t} = -\frac{\partial}{\partial t}\left(\frac{\vec{n}}{v}\cdot\vec{E}\right) \qquad \nabla\cdot\vec{B} = \nabla(t-\frac{\vec{r}\cdot\vec{n}}{v})\cdot\frac{\partial\vec{B}}{\partial t} = -\frac{\partial}{\partial t}\left(\frac{\vec{n}}{v}\cdot\vec{B}\right)$$

它们化为

$$\frac{\partial}{\partial t}(\vec{n}\cdot\vec{E}) = 0 \quad \rightarrow \quad \vec{n}\cdot\vec{E} = \text{时间无关?} \qquad \frac{\partial}{\partial t}(\vec{n}\cdot\vec{B}) = 0 \quad \rightarrow \quad \vec{n}\cdot\vec{B} = \text{时间无关项}$$

略去时间无关项(静态电磁场),则$\vec{E},\vec{B},\vec{n}$服从右手关系,且$vB = E$,

$$\begin{aligned} W_e &= \frac{1}{2}\epsilon E^2 = \frac{1}{2}\epsilon v^2B^2 = \frac{1}{2\mu}B^2 = W_m \\ \vec{S} &= \vec{E}\times\vec{H} = \vec{E}\times\frac{\vec{B}}{\mu} = -(\vec{v}\times\vec{B})\times\frac{\vec{B}}{\mu} = \vec{v}\frac{B^2}{\mu} = \vec{v}(W_e+W_m) \end{aligned}$$

§3.2 定态波动方程及平面波解

- 导体

前面讨论了当$\rho_f = 0, \vec{j}_c = 0$时在理想绝缘介质中传播的平面电磁波. 现在我们将讨论扩充到可以包括导体的情形. 导体与绝缘介质的差别在于对导体存在欧姆定律$\vec{j} = \gamma\vec{E}$. 我们前面讨论的$\vec{j} = 0$时的绝缘介质也可以包括在这个定律中,这时只需取$\gamma = 0$. 由于电荷守恒,

$$\frac{\partial\rho}{\partial t} = -\nabla\cdot\vec{j} = -\nabla\cdot(\gamma\vec{E}) = -\frac{\gamma}{\epsilon}\nabla\cdot\vec{D} = -\frac{\gamma}{\epsilon}\rho$$

它的解为

$$\rho(t) = \rho(0)e^{-\frac{\gamma}{\epsilon}t}$$

即在导体中,电荷密度是很快衰减的(在静电时,导体内部不带电,电荷只分布在表面上). 我们只讨论如下两种情况

$$\begin{cases} \gamma\sim 0 & \rho(0) = 0 \\ \gamma \gg \epsilon\omega & \omega: \text{电磁波的角频率} \end{cases}$$

在这两种情况下,我们可以近似地取$\rho(t) = 0$.

- 定态电磁波

  对一般的介质,物质电磁性质方程,$\vec{D}=\epsilon\vec{E},\vec{B}=\mu\vec{H},\vec{j}=\gamma\vec{E}$在迅变的电磁场下不成立.但对以一定频率$\omega$作振荡的电磁波,上述方程仍可成立,即

  $$\vec{D}(\omega)=\epsilon(\omega)\vec{E}(\omega)\qquad\vec{B}(\omega)=\mu(\omega)\vec{H}(\omega)\qquad\vec{j}(\omega)=\gamma(\omega)\vec{E}(\omega)$$

  其中,介电常数$\epsilon(\omega)$,磁导率$\mu(\omega)$,电导率$\gamma(\omega)$一般说是频率的函数.这种现象叫介质的色散. 以一定频率作振荡的电磁波叫定态电磁波(或叫单色波).

  在一般情况下,即使电磁波不是单色波,也可用傅里叶(Fourier)分析(或叫频谱分析)方法将其分解为不同频率的单色波的迭加.

  以下只讨论定态电磁波,设其角频率为ω,电磁场对时间的依赖关系为$\cos\omega t$. 为方便,我们经常用复数形式:

  $$\vec{E}(\vec{r},t)=\vec{E}(\vec{r})e^{-i\omega t}\qquad\vec{B}(\vec{r},t)=\vec{B}(\vec{r})e^{-i\omega t}$$

  注意

  - ω必须取实数,否则$e^{-i\omega t}$不再是振荡因子,变成随时间的增强(非物理)或衰减因子,形不成持久的波动.
  - 物理的电磁场是$\vec{E}(\vec{r},t),\vec{B}(\vec{r},t)$的实部.
  - 不要将抽出时间因子$e^{-i\omega t}$以后的电场强度与$\vec{E}(\vec{r},t)$混淆.

  将上面表达式代入麦克斯韦方程组,约去共同因子$e^{-i\omega t}$,得

  $$\begin{cases}\nabla\times\vec{E}=i\omega\vec{B}\\ \nabla\times\vec{H}=\vec{j_c}-i\omega\vec{D}\\ \nabla\cdot\vec{D}=0\\ \nabla\cdot\vec{B}=0\end{cases}\quad\text{代入物质电磁方程}\rightarrow\quad\begin{cases}\nabla\times\vec{E}=i\omega\vec{B}\\ \nabla\times\vec{B}=\mu(\gamma-i\omega\epsilon)\vec{E}\\ \nabla\cdot\vec{E}=0\\ \nabla\cdot\vec{B}=0\end{cases}$$

  因此

  $$\nabla^2\vec{E}=\nabla^2\vec{E}-\nabla(\nabla\cdot\vec{E})=-\nabla\times(\nabla\times\vec{E})=-i\omega\nabla\times\vec{B}=i\omega\mu(\gamma-i\omega\epsilon)\vec{E}$$
  $$\nabla^2\vec{B}=\nabla^2\vec{B}-\nabla(\nabla\cdot\vec{B})=-\nabla\times(\nabla\times\vec{B})=\mu(\gamma-i\omega\epsilon)\nabla\times\vec{E}=i\omega\mu(\gamma-i\omega\epsilon)\vec{B}$$

  我们得到

  $$\begin{cases}(\nabla^2+k^2)\vec{E}=0\\ \nabla\cdot\vec{E}=0\\ \vec{B}=-\frac{i}{\omega}\nabla\times\vec{E}\end{cases}\quad\begin{matrix}\vec{E},\vec{B}\text{相互不独立}\\ \longleftarrow\!\!-\!\!-\!\!-

$\vec{k}$叫波矢, 将其实虚部明显写出$\vec{k}=\vec{k}_R+i\vec{k}_I$.
$\vec{E}_0,\vec{B}_0$叫复振幅,将其模和幅角明显写出$E_{0n}=\|E_{0n}\|e^{i\delta_{En}}$, $B_{0n}=\|B_{0n}\|e^{i\delta_{Bn}}$. 其中,$\|E_{0n}\|$代表对$E_{0n}$求复数模. 结合起来,

$$\vec{E}(\vec{r},t)=\sum_{n=1}^{3}\vec{e}_n\|E_{0n}\|e^{-\vec{k}_I\cdot\vec{r}}e^{i(\vec{k}_R\cdot\vec{r}-\omega t+\delta_{En})},\quad \vec{B}(\vec{r},t)=\sum_{n=1}^{3}\vec{e}_n\|B_{0n}\|e^{-\vec{k}_I\cdot\vec{r}}e^{i(\vec{k}_R\cdot\vec{r}-\omega t+\delta_{Bn})}$$

其中:
$\|E_{0n}\|e^{-\vec{k}_I\cdot\vec{r}}$, $\|B_{0n}\|e^{-\vec{k}_I\cdot\vec{r}}$分别是电场强度和磁感应强度在$n$方向分量的振幅.
$\vec{k}_R\cdot\vec{r}-\omega t+\delta_{En}$, $\vec{k}_R\cdot\vec{r}-\omega t+\delta_{Bn}$分别是电场强度和磁感应强度在$n$方向分量的位相.
这样的解具有如下性质

- 在给定时刻t,等相面是与$\vec{k}_R$垂直的平面. $\vec{k}_R\cdot\vec{r}=\vec{k}_R\cdot(\vec{r}+\vec{r}_\perp)\quad \vec{r}_\perp\cdot\vec{k}_R=0.$
- 等相面以速率$v_\phi=\omega/k_R$沿$\vec{k}_R$方向运动.

$$-\omega t_1+\vec{k}_R\cdot\vec{r}_1=-\omega t_2+\vec{k}_R\cdot\vec{r}_2\qquad\rightarrow\qquad 1=\frac{\vec{k}_R}{\omega}\cdot\frac{(\vec{r}_2-\vec{r}_1)}{t_2-t_1}\frac{\vec{k}_R}{\omega}\cdot\vec{v}_\phi$$

- 当$\vec{k}_I=0$时,等相面就是等值面.
- 当$\vec{k}_I\neq 0$时,等值区域由$\vec{k}_R\cdot\vec{r}=$常数和$\vec{k}_I\cdot\vec{r}=$常数联合求解决定,它一般是一条线.振幅最大处位于$\vec{k}_I\cdot\vec{r}=0$处. 偏离最大振幅处为衰减的振幅.
- 波矢
  $\vec{k}_R$的方向反映的是位相的传播方向. $\vec{k}_I$的方向反映的是振幅的最大衰减方向. $k^2=\omega^2\mu\epsilon+i\omega\mu\gamma=\vec{k}\cdot\vec{k}=k_R^2-k_I^2+2i\vec{k}_R\cdot\vec{k}_I$给出

  $$k_R^2-k_I^2=\mu\epsilon\omega^2\qquad\qquad k_Rk_I\cos\theta=\frac{1}{2}\omega\mu\gamma$$

  其中$\cos\theta\equiv\frac{\vec{k}_R}{k_R}\cdot\frac{\vec{k}_I}{k_I}$, 若$\omega\mu\gamma>0$,则$0<\theta<\pi/2$.
  k_I代表了电磁波振幅的衰减.从另一个角度,它反映了物质对电磁波的吸收. k_I大,吸收强,k_I小,吸收弱. 注意对绝缘介质,$\gamma=0,k_I=0$, 因此对绝缘介质,电磁波无吸收,电磁能量不耗散.通常将波幅降至原值$1/e$的传播距离称为穿透深度$\delta=\frac{1}{k_I\cos\theta}$.
  k_I在实际应用中有重要意义.对一种确定物质, k_I一般与ω有关, 对某些ω值较大,对另外的ω值较小,它反映这种物质对某些频率的电磁波吸收较强,而对其它频率的电磁波吸收较弱.如,人之所以只能看到可见光范围的电磁波,就是因为视网膜的k_I在可见光范围内较大,而对其它频率很小.另外,大气对很大一部分频率的电磁波吸收很强,除可见光外,只对一小部分频段的电磁波吸收较弱,称为大气的窗户.大气层的这个性质好处是保护地球上的动植物免受来自宇宙太空的电磁射线的损害,坏处是限制了人类获取来自宇宙太空的信息,在地球上,只能接收几个窗口频段的电磁波.
- 振幅
  复振幅$\vec{E}_0,\vec{B}_0$满足

  $$\vec{k}\cdot\vec{E}_0=0\qquad\qquad \vec{B}_0=\frac{\vec{k}}{\omega}\times\vec{E}_0$$

  若$k_I=0$(绝缘介质就是这种情形), 上式意味$\vec{k},\vec{E}_0,\vec{B}_0$成右手关系,且$B_0=\frac{k}{\omega}E_0$, $\vec{B}$与$\vec{E}$同位相.
  若$k_i\neq 0$, 右手关系一般不成立,$\vec{B}$与$\vec{E}$一般也不同位相. 若将$\vec{E}_0,\vec{B}_0$的实虚部明显写出来,

  $$\vec{E}_0=\vec{E}_{0R}+i\vec{E}_{0I}\qquad\qquad \vec{B}_0=\vec{B}_{0R}+i\vec{B}_{0I}$$

则对实虚部的约束关系可写为

$$\vec{k}_R \cdot \vec{E}_{0R} - \vec{k}_I \cdot \vec{E}_{0I} = 0 \qquad \vec{k}_R \cdot \vec{E}_{0I} + \vec{k}_I \cdot \vec{E}_{0R} = 0$$
$$\vec{B}_{0R} = \frac{\vec{k}_R}{\omega} \times \vec{E}_{0R} - \frac{\vec{k}_I}{\omega} \times \vec{E}_{0I} \qquad \vec{B}_{0I} = \frac{\vec{k}_R}{\omega} \times \vec{E}_{0I} + \frac{\vec{k}_I}{\omega} \times \vec{E}_{0R}$$

对比前面引入的$\vec{E}_0, \vec{B}_0$的复数模和相角关系,

$$\vec{E}_{0R} = \sum_{n=1}^{3} \|E_{0n}\| \vec{e}_n \cos\delta_{En} \qquad \vec{E}_{0I} = \sum_{n=1}^{3} \|E_{0n}\| \vec{e}_n \sin\delta_{En}$$
$$\vec{B}_{0R} = \sum_{n=1}^{3} \|B_{0n}\| \vec{e}_n \cos\delta_{Bn} \qquad \vec{B}_{0I} = \sum_{n=1}^{3} \|B_{0n}\| \vec{e}_n \sin\delta_{Bn}$$

– 偏振

在$\vec{k}$为实的情况下,取z轴为$\vec{k}$方向(即电磁波沿z轴正方向传播),则

$$E_{0z} = 0 \qquad \vec{E}(\vec{r}, t) = \|E_{0x}\| e^{i(kz - \omega t + \delta_{Ex})} \vec{e}_1 + \|E_{0y}\| e^{i(kz - \omega t + \delta_{Ey})} \vec{e}_2$$

在某一z取定的xy平面上,电场强度矢量随时间不断变化,可以有如下行为

$$\begin{cases} \delta_{Ex} - \delta_{Ey} = 0, \pi & \text{线偏振} \\ \|E_{0x}\| = \|E_{0y}\| & \begin{cases} \delta_{Ex} - \delta_{Ey} = \frac{\pi}{2} & \text{左旋} \\ \delta_{Ex} - \delta_{Ey} = \frac{3\pi}{2} & \text{右旋} \\ \delta_{Ex} - \delta_{Ey} \text{不固定} & \text{自然光} \end{cases} \\ \text{一般: 椭圆} \end{cases}$$

在实际应用中,有些晶体对某一特定方向上的电磁振动有强烈的吸收,一个椭圆偏振或自然光通过它会变成线偏振光. 立体电影即是利用这种性质使双眼看到不同的影象, 产生立体效果.

– 能量密度, 能流密度

$$W_e = \frac{1}{2}\epsilon \vec{E}_R \cdot \vec{E}_R \qquad W_m = \frac{1}{2\mu} \vec{B}_R \cdot \vec{B}_R \qquad \vec{S} = \frac{1}{\mu} \vec{E}_R \times \vec{B}_R$$

注意

$$\vec{E}_R \cdot \vec{E}_R = \sum_{n=1}^{3} E_{Rn} E_{Rn} = \sum_{n=1}^{3} \|E_{0n}\|^2 e^{-2\vec{k}_I \cdot \vec{r}} \cos^2(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{En})$$
$$\vec{B}_R \cdot \vec{B}_R = \sum_{n=1}^{3} B_{Rn} B_{Rn} = \sum_{n=1}^{3} \|B_{0n}\|^2 e^{-2\vec{k}_I \cdot \vec{r}} \cos^2(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{Bn})$$
$$\vec{E}_R \times \vec{B}_R = \sum_{n,m,p=1}^{3} \epsilon_{nmp} E_{Rn} B_{Rm} \vec{e}_p = \sum_{n,m,p=1}^{3} \|E_{0n}\| \|B_{0m}\| e^{-2\vec{k}_I \cdot \vec{r}} \vec{e}_p \cos(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{En}) \cos(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{Bn})$$

并且

$$\begin{aligned} \overline{\cos^2(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{En})} &= \frac{1}{2}\overline{[1 + \cos 2(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{En})]} = \frac{1}{2} \\ \overline{\cos^2(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{Bn})} &= \frac{1}{2}\overline{[1 + \cos 2(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{Bn})]} = \frac{1}{2} \\ \overline{\cos(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{En}) \cos(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{Bn})} &= \frac{1}{2}\overline{[\cos(2\vec{k}_R \cdot \vec{r} - 2\omega t + \delta_{En} + \delta_{Bn}) + \cos(\delta_{En} - \delta_{Bn})]} \\ &= \frac{1}{2}\cos(\delta_{En} - \delta_{Bn}) \end{aligned}$$

因此

$$\overline{\vec{E}_R\cdot\vec{E}_R}=\frac{1}{2}\sum_{n=1}^{3}\|E_{0n}^2\|e^{-2\vec{k}_I\cdot\vec{r}}=\frac{1}{2}Re(\vec{E}\cdot\vec{E}^*)\qquad \overline{\vec{B}_R\cdot\vec{B}_R}=\frac{1}{2}\sum_{n=1}^{3}\|B_{0n}^2\|e^{-2\vec{k}_I\cdot\vec{r}}=\frac{1}{2}Re(\vec{B}\cdot\vec{B}^*)$$

$$\overline{\vec{E}_R\times\vec{B}_R}=\frac{1}{2}\sum_{n,m,p=1}^{3}\epsilon_{nmp}\|E_{0n}\|\|B_{0m}\|\vec{e}_p e^{-2\vec{k}_I\cdot\vec{r}}\cos(\delta_{En}-\delta_{Bn})=\frac{1}{2}Re(\vec{E}\times\vec{B}^*)$$

最后

$$\begin{aligned}\overline{W_e}&=\frac{1}{2}\epsilon\overline{\vec{E}_R\cdot\vec{E}_R}=\frac{1}{4}\epsilon Re(\vec{E}\cdot\vec{E}^*)\\ \overline{W_m}&=\frac{1}{2\mu}\overline{\vec{B}_R\cdot\vec{B}_R}=\frac{1}{4\mu}Re(\vec{B}\cdot\vec{B}^*)=\frac{1}{4\mu}Re\Big[(\frac{\vec{k}}{\omega}\times\vec{E})\cdot(\frac{\vec{k}^*}{\omega}\times\vec{E}^*)\Big]\\ &=\frac{1}{4\mu\omega^2}Re\Big[(\vec{k}\cdot\vec{k}^*)(\vec{E}\cdot\vec{E}^*)-(\vec{k}\cdot\vec{E}^*)(\vec{k}^*\cdot\vec{E})\Big]\end{aligned}$$

对绝缘体,$\vec{k}=\vec{k}^*,\vec{k}\cdot\vec{E}=0\rightarrow$, $W_e=W_m$,但导体则$W_e\neq W_m$,即电能和磁能是等量分布在绝缘体的电磁波中,但不等量在导体中.

$$\begin{aligned}\overline{\vec{S}}&=\frac{\overline{\vec{E}_R\times\vec{B}_R}}{\mu}=\frac{1}{2\mu}Re(\vec{E}\times\vec{B}^*)=\frac{1}{2\mu}Re(\vec{E}^*\times\vec{B})=\frac{1}{2\mu}Re[\vec{E}^*\times(\frac{\vec{k}}{\omega}\times\vec{E})]\\ &=\frac{1}{2\mu\omega}Re[\vec{k}(\vec{E}\cdot\vec{E}^*-\vec{E}(\vec{k}\cdot\vec{E}^*)]\end{aligned}$$

对绝缘体,能量沿波传播方向流动,对导体则还有其它方向的流动.

§3.3 电磁波在界面上的反射和折射

由于电磁波在金属中衰减的很历害,我们只讨论电磁波从一绝缘介质入射到另一绝缘或导电介质中,电磁波的行为. 取ϵ,μ,γ都为实数.

- 边界条件

  设界面位于$z=0$的xy平面.入射电磁波方向位于xz平面,与z轴夹角θ.入射波为频率ω的平面电磁波.

$\vec{A}_1=\vec{A}_{10}e^{i(\vec{k}_1\cdot\vec{r}-\omega t)}$ $\leftarrow$ 入射波; $\vec{A}_2=\vec{A}_{20}e^{i(\vec{k}_2\cdot\vec{r}-\omega_2 t)}$ $\leftarrow$ 透射波; $\vec{A}_3=\vec{A}_{30}e^{i(\vec{k}_3\cdot\vec{r}-\omega_3 t)}$ $\leftarrow$ 反射波

$\vec{A}\subset(\vec{E},\vec{H})$

$$\vec{k}_1=k_1(\sin\theta_1\vec{e}_1+\cos\theta_1\vec{e}_3)\qquad \vec{k}_3=k_{3x}\vec{e}_1+k_{3y}\vec{e}_2+k_{3z}\vec{e}_3\qquad \vec{k}_2=\vec{k}_{2R}+i\vec{k}_{2I}$$

$$k_1=\frac{\omega}{v}=\sqrt{\mu_1\epsilon_1}\omega\qquad k_3=\frac{\omega_3}{v}=\sqrt{\mu_1\epsilon_1}\omega_3\qquad \begin{cases}k_R^2-k_I^2=\mu_2\epsilon_2\omega_2^2\\ k_{2R}k_{2I}\cos\theta=\frac{1}{2}\omega_2\mu_2\gamma_2\end{cases}$$

已知:$\vec{k}_1,\omega,\vec{A}_{10},\mu_1,\epsilon_1,\gamma_1,\mu_2,\epsilon_2,\gamma_2,\theta_1$, 我们需要求:$\vec{k}_2,\vec{k}_3,\omega_2,\omega_3,\vec{A}_{20},\vec{A}_{30}$. 出发点是边条件$\vec{n}\times(\vec{E}_2-\vec{E}_1)=0,\vec{n}\times(\vec{H}_2-\vec{H}_1)=\vec{i}_c,\vec{n}\cdot(\vec{D}_2-\vec{D}_1)=\sigma_f,\vec{n}\cdot(\vec{B}_2-\vec{B}_1)=0$.上述边条件只有两个是独立的. 我们讨论$\vec{i}_c=0,\sigma_f=0$的情况. 选$\vec{H},\vec{E}$的边条件,即$\vec{n}\times(\vec{A}_1+\vec{A}_3)\Big|_{z=0}=\vec{n}\times\vec{A}_2\Big|_{z=0}$或

$$\vec{n}\times[\vec{A}_{10}e^{i(k_{1x}x+k_{1y}y-\omega t)}+\vec{A}_{30}e^{i(k_{3x}x+k_{3y}y-\omega_3 t)}]=\vec{n}\times\vec{A}_{20}e^{i(k_{2x}x+k_{2y}y-\omega_2 t)}$$

这个方程的解为

$$\begin{cases} \omega = \omega_2 = \omega_3 \\ k_{1x} = k_{2x} = k_{3x} = k_1 \sin\theta_1 = \sqrt{\mu_1\epsilon_1}\sin\theta_1 \\ k_{1y} = k_{2y} = k_{3y} = 0 \\ \vec{n} \times (\vec{A}_{10} + \vec{A}_{30}) = \vec{n} \times \vec{A}_{20} \end{cases}$$

- 反射透射波的波矢
  - 反射波

$$\begin{array}{ccccc} \omega_3 = \omega, & & k_3 = k_1, & & \vec{k}_3 = k_3\sin\theta_3\vec{e}_1 - k_3\cos\theta_3\vec{e}_3 \\ k_{3x} = k_{1x} & \rightarrow & k_3\sin\theta_3 = k_1\sin\theta_1 & \rightarrow & \theta_3 = \theta_1 \quad \text{反射定律} \end{array}$$

    所以,反射波也是频率为ω的平面单色波,位于入射波,且入射角等于反射角.
  - 透射波

$$\omega_2 = \omega, \qquad k_{2Rx} = \sqrt{\mu_1\epsilon_1}\omega\sin\theta_1, \qquad k_{2Ix} = 0, \qquad k_{2Ry} = k_{2Iy} = 0$$
$$k_R^2 - k_I^2 = \mu_2\epsilon_2\omega_2^2, \qquad k_{2R}k_{2I}\cos\theta = \frac{1}{2}\omega_2\mu_2\gamma_2$$

    结合起来

$$k_{2Rz}^2 - k_{2Iz}^2 = k_{2R}^2 - k_{2Rx}^2 - k_{2Ry}^2 - k_{2I}^2 + k_{2Ix}^2 + k_{2Iy}^2 = \mu_2\epsilon_2\omega^2 - \mu_1\epsilon_1\omega^2\sin^2\theta_1$$
$$k_{2Rz}k_{2Iz} = \vec{k}_{2R}\cdot\vec{k}_{2Iz} - k_{2Rx}k_{2Ix} - k_{2Ry}k_{2Iy} = \frac{1}{2}\omega\mu_2\gamma_2$$

    最后解出

$$\begin{aligned} k_{2Rx} &= \sqrt{\mu_1\epsilon_1}\omega\sin\theta_1 \\ k_{2Ry} &= 0 \\ k_{2Rz} &= \frac{1}{\sqrt{2}}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1)\omega^2 + \sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1)^2\omega^4 + \mu_2\gamma_2\omega^2}} \\ k_{2Ix} &= 0 \\ k_{2Iy} &= 0 \\ k_{2Iz} &= \frac{\omega\mu_2\gamma_2}{\sqrt{2}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1)\omega^2 + \sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1)^2\omega^4 + \mu_2\gamma_2\omega^2}}} \end{aligned}$$

    透射波也是频率为ω的单色波,位于入射面,衰减方向垂直于界面(z方向).
    讨论
    1. 绝缘透射介质: $\gamma_2 = 0$, 折射率$n \equiv \sqrt{\mu_r\epsilon_r} = c\sqrt{\mu\epsilon}$

$$k_2 = \sqrt{\mu_2\epsilon_2}\omega = \frac{n_2}{c}\omega \qquad \frac{k_2}{k_1} = \frac{n_2}{n_1}$$
$$k_{2Rz} = \frac{\omega}{\sqrt{2}}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1) + \|\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1\|}$$
$$k_{2Iz} = \frac{\omega\mu_2\gamma_2}{\sqrt{2}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1) + \|\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1\|}}$$

(a) $\mu_2\epsilon_2 \geq \mu_1\epsilon_1 \qquad \rightarrow \qquad n_2 \geq n_1$,即光疏→光密

$$k_{2Rz} = \omega\sqrt{\mu_2\epsilon_2 - \mu_1\epsilon_1 \sin^2\theta_1} \qquad k_{2Iz} = 0 \qquad \text{无衰减}$$
$$\vec{k}_2 = k_2 \sin\theta_2 \vec{e}_1 + k_2 \cos\theta_2 \vec{e}_3 = \omega\sqrt{\mu_2\epsilon_2}(\sin\theta_2\vec{e}_1 + \cos\theta_2\vec{e}_3)$$
$$\rightarrow \quad \omega\sqrt{\mu_2\epsilon_2}\sin\theta_2 = \omega\sqrt{\mu_1\epsilon_1}\sin\theta_1 \quad \rightarrow \quad n_2\sin\theta_2 = n_1\sin\theta_1 \qquad \text{折射定律}$$

注意,这时$\sin\theta_2 = \frac{n_1}{n_2}\sin\theta_1 \leq 1$

(b) $\mu_2\epsilon_2 < \mu_1\epsilon_1 \qquad \sin\theta_1 \leq \sqrt{\frac{\mu_2\epsilon_2}{\mu_1\epsilon_1}} = \frac{n_2}{n_1}$, 即光密→光疏.

公式同前,$\vec{k}_2$仍为实, $n_2\sin\theta_2 = n_1\sin\theta_1$注意,这时仍有$\sin\theta_2 = \frac{n_1}{n_2}\sin\theta_1 \leq 1$

(c) $\mu_2\epsilon_2 < \mu_1\epsilon_1 \qquad \sin\theta_1 > \sqrt{\frac{\mu_2\epsilon_2}{\mu_1\epsilon_1}} = \frac{n_2}{n_1}$, 即光密→光疏.

$$k_{2Rz} = 0$$
$$k_{2Iz} = \left.\frac{\omega\mu_2\gamma_2}{\sqrt{2}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1)\omega^2 + (\mu_1\epsilon_1\sin^2\theta_1 - \mu_2\epsilon_2)\omega^2[1 + \frac{\mu_2\gamma_2^2\omega^2}{2(\mu_1\epsilon_1\sin^2\theta_1 - \mu_2\epsilon_2)^2\omega^4}]}}\right|_{\gamma_2\rightarrow 0}$$
$$= \omega\sqrt{\mu_1\epsilon_1\sin^2\theta_1 - \mu_2\epsilon_2}$$

这时

$$\vec{k}_{2R} = \sqrt{\mu_1\epsilon_1}\omega\sin\theta_1\vec{e}_1 \qquad v_{2\phi} = v_{2\phi,x} = \frac{\omega}{k_R} = \frac{1}{\sqrt{\mu_1\epsilon_1}\sin\theta_1} = \frac{v_1}{\sin\theta_1}$$
$$\vec{k}_{2I} = \omega\sqrt{\mu_1\epsilon_1\sin^2\theta_1 - \mu_2\epsilon_2}\vec{e}_3$$

波沿x方向贴界面传播,衰减方向为z方向.这种情况叫全反射.(透射波不向介质内传播)

$$\vec{A}_2 = \vec{A}_{20}e^{-\omega z\sqrt{\mu_1\epsilon_1\sin\theta_1 - \mu_2\epsilon_2}}e^{i(\sqrt{\mu_1\epsilon_1}\omega\sin\theta_1 x - \omega t)}$$

2. 导电透射介质: $\gamma \neq 0$

$$\vec{A}_2 = \vec{A}_{20}e^{-k_{2Iz}z}e^{i(k_{2Rx}x + k_{2Rz}z - \omega t)}$$

* 透射波等相面沿$\vec{k}_{2R} = k_{2Rx}\vec{e}_1 + k_{2Rz}\vec{e}_3$方向传播, 频率$\omega$,速率$v = \frac{\omega}{k_{2R}} = \frac{\omega}{\sqrt{k_{2Rx}^2 + k_{2RZ}^2}}$.
* 透射波有衰减,方向在z方向.

- 反射透射波的振幅

由前面的讨论知麦克斯韦方程组将给出对振幅的约束

$$\begin{cases} \vec{k}_3\cdot\vec{E}_{30} = 0 \\ \vec{k}_2\cdot\vec{E}_{20} = 0 \end{cases} \rightarrow \begin{cases} k_{3x}E_{30x} + k_{3z}E_{30z} = 0 \\ k_{2x}E_{20x} + k_{2z}E_{20z} = 0 \end{cases} \rightarrow \begin{cases} E_{30z} = -\frac{k_{3x}}{k_{3z}}E_{30x} \\ E_{20z} = -\frac{k_{2x}}{k_{2z}}E_{20x} \end{cases} \qquad \begin{cases} \vec{B}_{30} = \frac{\vec{k}_3}{\omega}\times\vec{E}_{30} \\ \vec{B}_{20} = \frac{\vec{k}_2}{\omega}\times\vec{E}_{20} \end{cases}$$

为方便,分两种情况讨论:一种入射波电场位于入射面,磁场垂直入射面.另一种入射波电场垂直入射面,磁场位于入射面.一般情况总可以化为这两种情况的迭加.

1. 入射波电场位于入射面

入射波振幅

$$\vec{E}_{10} = E_{10}(\vec{e}_1\cos\theta_1 - \vec{e}_3\sin\theta_1) \qquad \vec{B}_{10} = \frac{\vec{k}_1}{\omega}\times\vec{E}_{10} = \frac{k_1}{\omega}E_{10}\vec{e}_y$$

由边条件

$$\vec{E}_{10t}+\vec{E}_{30t}=\vec{E}_{20t}\qquad\qquad \vec{H}_{10t}+\vec{H}_{30t}=\vec{H}_{20t}$$

它具体化为

$$\begin{cases} E_{30y}=E_{20y} & \clubsuit \\ \frac{1}{\mu_1}B_{30x}=\frac{1}{\mu_2}B_{20x} & \diamondsuit \end{cases}\qquad\qquad \begin{cases} E_{10}\cos\theta_1+E_{30x}=E_{20x} & \spadesuit \\ \frac{1}{\mu_1}\frac{k_1}{\omega}E_{10}+\frac{1}{\mu_1}B_{30y}=\frac{1}{\mu_2}B_{20y} & \heartsuit \end{cases}$$

结合起来

$$\diamondsuit\to\frac{1}{\mu_1}(\frac{\vec{k}_1}{\omega}\times\vec{E}_{30})_x=\frac{1}{\mu_2}(\frac{\vec{k}_2}{\omega}\times\vec{E}_{20})_x\to\frac{1}{\mu_1}(k_{3y}E_{30z}-k_{3z}E_{30y})=\frac{1}{\mu_2}(k_{2y}E_{20z}-k_{2z}E_{20y})$$
$$\to\frac{1}{\mu_1}k_{3z}E_{30y}=\frac{1}{\mu_2}k_{2z}E_{20y}\xrightarrow{\clubsuit}E_{30y}=E_{20y}=0\xrightarrow{\diamondsuit}B_{30x}=B_{20x}=0$$

反射波和透射波电场仍位于入射面,磁场垂直于入射面. 定义

$$\vec{e}_{2\parallel}=\frac{k_{2z}}{k_2}\vec{e}_1-\frac{k_{2x}}{k_2}\vec{e}_3\qquad\qquad \vec{e}_{3\parallel}=\frac{k_{3z}}{k_3}\vec{e}_1-\frac{k_{3x}}{k_3}\vec{e}_3=-\cos\theta_1\vec{e}_1-\sin\theta_1\vec{e}_3$$

则

$$\vec{E}_{20}=E_{20x}\vec{e}_1+E_{20z}\vec{e}_3=E_{20x}\vec{e}_1-\frac{k_{2x}}{k_{2z}}\vec{e}_3E_{20x}=\vec{e}_{2\parallel}\frac{k_2}{k_{2z}}E_{20x}$$
$$\vec{E}_{30}=E_{30x}\vec{e}_1+E_{30z}\vec{e}_3=E_{30x}\vec{e}_1-\frac{k_{3x}}{k_{3z}}\vec{e}_3E_{30x}=\vec{e}_{3\parallel}\frac{k_3}{k_{3z}}E_{30x}=-\vec{e}_{3\parallel}\frac{1}{\cos\theta_1}E_{30x}$$

因此

$$\heartsuit\to\frac{1}{\mu_2}(\frac{\vec{k}_2}{\omega}\times\vec{E}_{20})_y-\frac{1}{\mu_1}(\frac{\vec{k}_3}{\omega}\times\vec{E}_{30})_y=\frac{1}{\mu_1}\frac{k_1}{\omega}E_{10}$$
$$\to\frac{1}{\mu_2}(k_{2z}E_{20x}-k_{2x}E_{20z})-\frac{1}{\mu_1}(k_{3z}E_{30x}-k_{3x}E_{30z})=\frac{k_1}{\mu_1}E_{10}$$
$$\to\frac{1}{\mu_2}(k_{2z}+\frac{k_{2x}^2}{k_{2z}})E_{20x}-\frac{1}{\mu_1}(k_{3z}+\frac{k_{3x}^2}{k_{3z}})E_{30x}=\frac{1}{\mu_1}k_1E_{10}$$
$$\to\frac{1}{\mu_2}\frac{k_2^2}{k_{2z}}E_{20x}-\frac{1}{\mu_1}\frac{k_3^2}{k_{3z}}E_{30x}=\frac{1}{\mu_1}k_1E_{10}$$
$$+\spadesuit\to\qquad\frac{\mu_1}{\mu_2}\frac{k_2^2}{k_1k_{2z}}E_{20x}+\frac{1}{\cos\theta_1}E_{30x}=E_{10}\qquad\qquad E_{20x}-E_{30x}=E_{10}\cos\theta_1$$
$$\to E_{20x}=\frac{2\cos\theta_1}{1+\frac{\mu_1k_2^2\cos\theta_1}{\mu_2k_1k_{2z}}}E_{10}\qquad\qquad E_{30x}=\frac{1-\frac{\mu_1k_2^2\cos\theta_1}{\mu_2k_1k_{2z}}}{1+\frac{\mu_1k_2^2\cos\theta_1}{\mu_2k_1k_{2z}}}\cos\theta_1E_{10}$$

最后得到

$$\vec{E}_{20\parallel}=\frac{2\cos\theta_1}{\frac{k_{2z}}{k_2}+\frac{k_2\mu_1}{k_1\mu_2}\cos\theta_1}E_{10\parallel}\vec{e}_{2\parallel}\qquad\qquad \vec{E}_{30\parallel}=\frac{-\frac{k_{2z}}{k_2}+\frac{k_2\mu_1}{k_1\mu_2}\cos\theta_1}{\frac{k_{2z}}{k_2}+\frac{k_2\mu_1}{k_1\mu_2}\cos\theta_1}E_{10\parallel}\vec{e}_{3\parallel}$$

特别地,若$\gamma_2=0,\mu_2=\mu_1$,则:$k_{2z}=k_2\cos\theta_2 k_{2x}=k_2\sin\theta_2\to\vec{e}_{2\parallel}=\cos\theta_2\vec{e}_1-\sin\theta_2\vec{e}_3$, 且$k_2/k_1=n_2/n_1=\sin\theta_1/\sin\theta_2$,因此

$$\vec{E}_{20\parallel}=\frac{2\cos\theta_1}{\cos\theta_2+\frac{\sin\theta_1}{\sin\theta_2}\cos\theta_1}E_{10\parallel}\vec{e}_{2\parallel}=\frac{2\cos\theta_1\sin\theta_2}{\frac{1}{2}\sin2\theta_2+\frac{1}{2}\sin2\theta_1}E_{10\parallel}\vec{e}_{2\parallel}=\frac{2\cos\theta_1\sin\theta_2}{\sin(\theta_1+\theta_2)\cos(\theta_1+\theta_2)}E_{10\parallel}\vec{e}_{2\parallel}$$
$$\vec{E}_{30\parallel}=\frac{-\sin\theta_2\cos\theta_2+\sin\theta_1\cos\theta_1}{\sin\theta_2\cos\theta_2+\sin\theta_1\cos\theta_1}E_{10\parallel}\vec{e}_{3\parallel}=\frac{-\sin2\theta_2+\sin2\theta_1}{\sin2\theta_2+\sin2\theta_1}E_{10\parallel}\vec{e}_{3\parallel}=\frac{\tan(\theta_1-\theta_2)}{\tan(\theta_1+\theta_2)}E_{10\parallel}\vec{e}_{3\parallel}$$

2. 入射波电场垂直入射面

入射波振幅

$$\vec{E}_{10}=E_{10}\vec{e}_2 \qquad\qquad \vec{B}_{10}=\frac{\vec{k}_1}{\omega}\times\vec{E}_{10}=\frac{k_1}{\omega}E_{10}(-\vec{e}_1\cos\theta_1+\vec{e}_3\sin\theta_1)$$

由边条件

$$\vec{E}_{10t}+\vec{E}_{30t}=\vec{E}_{20t} \qquad\qquad \vec{H}_{10t}+\vec{H}_{30t}=\vec{H}_{20t}$$

它具体化为

$$\begin{cases} E_{30x}=E_{20x} & \clubsuit \\ \frac{1}{\mu_1}B_{30y}=\frac{1}{\mu_2}B_{20y} & \diamondsuit \end{cases} \qquad\qquad \begin{cases} E_{10y}+E_{30y}=E_{20y} & \spadesuit \\ \frac{1}{\mu_1}(B_{10x}+B_{30x})=\frac{1}{\mu_2}B_{20x} & \heartsuit \end{cases}$$

计算过程见作业. 最后得到

$$\vec{E}_{20\perp}=\frac{2}{1+\frac{\mu_1k_{2z}}{\mu_2k_1\cos\theta_1}}E_{10\perp}\vec{e}_2 \qquad\qquad \vec{E}_{30\perp}=\frac{1-\frac{\mu_1k_{2z}}{\mu_2k_1\cos\theta_1}}{1+\frac{\mu_1k_{2z}}{\mu_2k_1\cos\theta_1}}E_{10\perp}\vec{e}_2$$

特别地,若$\gamma_2=0,\mu_2=\mu_1$,则:$k_{2z}=k_2\cos\theta_2=k_1\sin\theta_1\cos\theta_2/\sin\theta_2$因此

$$\vec{E}_{20\perp}=\frac{2}{1+\frac{\sin\theta_1\cos\theta_2}{\sin\theta_2\cos\theta_1}}E_{10\perp}\vec{e}_2=\frac{2\cos\theta_1\sin\theta_2}{\sin(\theta_1+\theta_2)}E_{10\perp}\vec{e}_2$$

$$\vec{E}_{30\perp}=\frac{1-\frac{\sin\theta_1\cos\theta_2}{\sin\theta_2\cos\theta_1}}{1+\frac{\sin\theta_1\cos\theta_2}{\sin\theta_2\cos\theta_1}}E_{10\perp}\vec{e}_2=\frac{\sin(\theta_2-\theta_1)}{\sin(\theta_2+\theta_1)}E_{10\perp}\vec{e}_2$$

3. 一般情况

$$\vec{E}_{10}=\vec{E}_{10\parallel}+\vec{E}_{10\perp}=E_{10\parallel}(\vec{e}_1\cos\theta_1-\vec{e}_3\sin\theta_1)+E_{10\perp}\vec{e}_2$$

$$\vec{E}_{20}=\vec{E}_{20\parallel}+\vec{E}_{20\perp}=E_{10\parallel}(\vec{e}_1\frac{k_{2z}}{k_2}-\vec{e}_3\frac{k_{2x}}{k_2})\frac{\cos\theta_1}{\frac{k_{2z}}{k_2}+\frac{\mu_1k_2}{\mu_2k_1}\cos\theta_1}+E_{10\perp}\vec{e}_2\frac{2}{1+\frac{\mu_1k_{2z}}{\mu_2k_1\cos\theta_1}}$$

$$\vec{E}_{30}=\vec{E}_{30\parallel}+\vec{E}_{30\perp}=E_{10\parallel}(\vec{e}_1\cos\theta_1+\vec{e}_3\sin\theta_1)\frac{-\frac{k_{2z}}{k_2}+\frac{\mu_1k_2}{\mu_2k_1}\cos\theta_1}{\frac{k_{2z}}{k_2}+\frac{\mu_1k_2}{\mu_2k_1}\cos\theta_1}+E_{10\perp}\vec{e}_2\frac{1-\frac{\mu_1k_{2z}}{\mu_2k_1\cos\theta_1}}{1+\frac{\mu_1k_{2z}}{\mu_2k_1\cos\theta_1}}$$

当透射介质是绝缘介质,对非全反射情形,$\vec{k}_2$为实,反透射波与入射波振幅之比为实,因此,反透射波与入射波同位相. 当透射介质是绝缘介质的非全反射情形或是导电介质的情形,$\vec{k}_2$有虚部, 反透射波与入射波振幅之比为复数,即,反透射波与入射波还相差位相,并且这个位相在反射波与透射波,$\parallel$波与$\perp$波之间是不同的,它将造成反射波,透射波,入射波偏振状态的差异.

- 反射透射波的能流

我们讨论能量流过界面的情况,利用

$$\overline{\vec{S}}=\frac{1}{2\mu\omega}Re[\vec{k}(\vec{E}\cdot\vec{E}^*)-\vec{E}(\vec{k}\cdot\vec{E}^*)]$$

– 对入射波

$$\overline{\vec{S}}_1=\frac{1}{2\mu_1\omega}Re[\vec{k}_1(\vec{E}_{10}\cdot\vec{E}^*_{10})]=\frac{1}{2\mu_1\omega}\vec{k}_1(\vec{E}_{10\parallel}\cdot\vec{E}^*_{10\parallel}+\vec{E}_{10\perp}\cdot\vec{E}^*_{10\perp})=\overline{\vec{S}}_{1\parallel}+\overline{\vec{S}}_{1\perp}$$

$$\overline{\vec{S}}_{1\parallel}=\frac{\vec{k}_1}{2\mu_1\omega}(\vec{E}_{10\parallel}\cdot\vec{E}^*_{10\parallel})=\frac{\vec{k}_1}{2\mu_1\omega}E_{10\parallel}E^*_{10\parallel}$$

$$\overline{\vec{S}}_{1\perp}=\frac{\vec{k}_1}{2\mu_1\omega}(\vec{E}_{10\perp}\cdot\vec{E}^*_{10\perp})=\frac{\vec{k}_1}{2\mu_1\omega}E_{10\perp}E^*_{10\perp}$$

其中利用了$\vec{k}_1=\vec{k}_1^*,\vec{k}_1\cdot\vec{E}_{10}=0$.

– 对反射波

$$\overline{\vec{S}}_3 = \frac{1}{2\mu_1\omega} Re[\vec{k}_3(\vec{E}_{30}\cdot\vec{E}^*_{30})] = \frac{1}{2\mu_1\omega}\vec{k}_3(\vec{E}_{30\parallel}\cdot\vec{E}^*_{30\parallel} + \vec{E}_{30\perp}\cdot\vec{E}^*_{30\perp}) = \overline{\vec{S}}_{3\parallel} + \overline{\vec{S}}_{3\perp}$$
$$\overline{\vec{S}}_{3\parallel} = \frac{\vec{k}_3}{2\mu_1\omega}(\vec{E}_{30\parallel}\cdot\vec{E}^*_{30\parallel}) = \frac{\vec{k}_3}{2\mu_1\omega}E_{30\parallel}E^*_{30\parallel}$$
$$\overline{\vec{S}}_{3\perp} = \frac{\vec{k}_3}{2\mu_1\omega}(\vec{E}_{30\perp}\cdot\vec{E}^*_{30\perp}) = \frac{\vec{k}_3}{2\mu_1\omega}E_{30\perp}E^*_{30\perp}$$

其中利用了$\vec{k}_3 = \vec{k}^*_3, \vec{k}_3\cdot\vec{E}_{30} = 0$.

– 对透射波

$$\begin{aligned}
\overline{\vec{S}}_2 &= \frac{1}{2\mu_2\omega} Re[\vec{k}_2(\vec{E}_2\cdot\vec{E}^*_2) - \vec{E}_2(\vec{k}_2\cdot\vec{E}^*_2)] \\
&= \frac{1}{2\mu_2\omega} Re[\vec{k}_2(\vec{E}_{2\parallel}\cdot\vec{E}^*_{2\parallel} + \vec{E}_{2\perp}\cdot\vec{E}^*_{2\perp}) - (\vec{E}_{2\parallel}+\vec{E}_{2\perp})(\vec{k}_2\cdot\vec{E}^*_{2\parallel})] = \overline{\vec{S}}_{2\parallel} + \overline{\vec{S}}_{2\perp} \\
\overline{\vec{S}}_{2\parallel} &= \frac{1}{2\mu_2\omega}\{\vec{k}_{2R}(\vec{E}_{2\parallel}\cdot\vec{E}^*_{2\parallel}) - Re(\vec{E}_{2\parallel}+\vec{E}_{2\perp})(\vec{k}_2\cdot\vec{E}_{2\parallel}\} \\
\overline{\vec{S}}_{2\perp} &= \frac{\vec{k}_{2R}}{2\mu_2\omega}(\vec{E}_{2\perp}\cdot\vec{E}^*_{2\perp})
\end{aligned}$$

注意,在$\overline{\vec{S}}_2$的公式中,无下标0.

取界面($z=0$)上一个小面元$\Delta\vec{S} = \Delta S\vec{e}_3$.

– 对⊥波定义透射系数

$$\begin{aligned}
D_\perp &\equiv \frac{\overline{\vec{S}}_{2\perp}\cdot\Delta\vec{S}}{\overline{\vec{S}}_{1\perp}\cdot\Delta\vec{S}}\Big|_{z=0} = \frac{\overline{S}_{2\perp z}}{\overline{S}_{1\perp z}}\Big|_{z=0} = \frac{\frac{k_{2zR}}{2\mu_2\omega}(\vec{E}_{2\perp}\cdot\vec{E}^*_{2\perp})}{\frac{k_{1z}}{2\mu_1\omega}(\vec{E}_{10\perp}\cdot\vec{E}^*_{10\perp})} = \frac{\mu_1 k_{2zR}}{\mu_2 k_1\cos\theta_1}\frac{\vec{E}_{20\perp}\cdot\vec{E}^*_{20\perp}}{E_{10\perp}E^*_{10\perp}}e^{-2\vec{k}_{2I}\cdot\vec{r}}\Big|_{z=0} \\
&= \frac{\mu_1 k_{2zR}}{\mu_2 k_1\cos\theta_1}\left\|\frac{2}{1+\frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}\right\|^2
\end{aligned}$$

– 对⊥波定义反射系数

$$R_\perp \equiv \frac{-\overline{\vec{S}}_{3\perp}\cdot\Delta\vec{S}}{\overline{\vec{S}}_{1\perp}\cdot\Delta\vec{S}}\Big|_{z=0} = -\frac{\overline{S}_{3\perp z}}{\overline{S}_{1\perp z}}\Big|_{z=0} = -\frac{\frac{k_{3z}}{2\mu_1\omega}(\vec{E}_{30\perp}\cdot\vec{E}^*_{30\perp})}{\frac{k_{1z}}{2\mu_1\omega}(\vec{E}_{10\perp}\cdot\vec{E}^*_{10\perp})} = \frac{\vec{E}_{30\perp}\cdot\vec{E}^*_{30\perp}}{E_{10\perp}E^*_{10\perp}} = \left\|\frac{1-\frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}{1+\frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}\right\|^2$$

能量守恒在界面上的体现为

$$\begin{aligned}
R_\perp + D_\perp &= \frac{\frac{4\mu_1 k_{2zR}}{\mu_2 k_1\cos\theta_1} + (1-\frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1})(1-\frac{\mu_1 k^*_{2z}}{\mu_2 k_1\cos\theta_1})}{\|1+\frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}\|^2} \\
&= \frac{\frac{2\mu_1}{\mu_2 k_1\cos\theta_1}(k_{2z}+k^*_{2z}) + 1 - \frac{\mu_1}{\mu_2 k_1\cos\theta_1}(k_{2z}+k^*_{2z}) + \frac{\mu_1^2 k_{2z}k^*_{2z}}{\mu_2^2 k_1^2\cos^2\theta_1}}{\|1+\frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}\|^2} = 1
\end{aligned}$$

即

$$(\overline{\vec{S}}_{2\perp} - \overline{\vec{S}}_{3\perp})\cdot\Delta\vec{S} = (D_\perp + R_\perp)\overline{\vec{S}}_{1\perp}\cdot\Delta\vec{S} = \overline{\vec{S}}_{1\perp}\cdot\Delta\vec{S}$$

从入射波流入ΔS的⊥电磁能量等于从ΔS流出的⊥的反射波和透射波电磁能量的迭加.

– 对∥波讨论见作业.

例: $\gamma_2 = 0$, $\mu_1 = \mu_2$, $k_2 = k_1 \sin\theta_1 / \sin\theta_2$, $k_{2z} = k_2 \cos\theta_2 = k_1 \sin\theta_1 \coth\theta_2$,

$$R_{\omega\parallel} = \left\| \frac{-\cos\theta_2 + \frac{\sin\theta_1}{\sin\theta_2}\cos\theta_1}{\cos\theta_2 + \frac{\sin\theta_1}{\sin\theta_2}\cos\theta_1} \right\|^2 = \left\| \frac{\sin 2\theta_1 - \sin 2\theta_2}{\sin 2\theta_1 + \sin 2\theta_2} \right\|^2 = \left\| \frac{\sin(\theta_1 - \theta_2)\cos(\theta_1 + \theta_2)}{\sin(\theta_1 + \theta_2)\cos(\theta_1 - \theta_2)} \right\|^2 = \frac{\tan^2(\theta_1 - \theta_2)}{\tan^2(\theta_1 + \theta_2)}$$

$$R_{\omega\perp} = \left\| \frac{-\sin\theta_1 \coth\theta_2 + \cos\theta_1}{\sin\theta_1 \coth\theta_2 + \cos\theta_1} \right\|^2 = \left[\frac{-\sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2}{\sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2} \right]^2 = \frac{\sin^2(\theta_1 - \theta_2)}{\sin^2(\theta_1 + \theta_2)}$$

$D_\parallel = 1 - R_\parallel$

$D_\perp = 1 - R_\perp$

注意

– R, D对$\theta_1 \leftrightarrow \theta_2$是对称的, 因而无论电磁波从哪边入射,$R, D$都是一样的.

– $R_{\omega\parallel}\Big|_{\theta_1+\theta_2=\frac{\pi}{2}} = 0$, 这样的解叫布儒斯特角.在气体激光管两端要放上两块倾斜的玻片,使激光的入射角为布儒斯特角. 这样对于电矢量垂直于入射面的偏振光将被衰减,而对电矢量位于入射面的偏振光可以来回无阻.用这样的方法可得到偏振激光.

- 光压

电磁波从真空射到物质表面的另一性质是使表面受力.单位表面受的力为

$$\begin{aligned} \vec{f} &= -\vec{n}\cdot \overrightarrow{\overrightarrow{\mathcal{J}}} = \vec{f}_E + \vec{f}_B \\ \vec{f}_E &= \frac{1}{2}\epsilon_0 E^2(\vec{n}\cos 2\theta_E + \vec{e}_E \sin 2\theta_E) \\ \vec{f}_B &= \frac{1}{2\mu_0} B^2(\vec{n}\cos 2\theta_B + \vec{e}_B \sin 2\theta_B) \end{aligned}$$

有时侯入射波的方向是不确定的,平均说它在所有方向上都等几率出现,因而平均贡献为

$$\overline{\cos 2\theta} = \frac{\int d\Omega \cos 2\theta}{\int d\Omega} = \frac{\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin\theta \cos 2\theta}{\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin\theta} = -\frac{1}{3}$$

$$\overline{\sin 2\theta} = \frac{\int d\Omega \sin 2\theta}{\int d\Omega} = \frac{\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin\theta \sin 2\theta}{\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin\theta} = 0$$

因此

$$\overline{\vec{f}} = -\frac{\vec{n}}{3}\left(\frac{1}{2}\epsilon_0 E^2 + \frac{1}{2}\mu_0 H^2\right) = -\frac{\vec{n}}{3} W$$

§3.4 谐振腔

- 方程及边条件

前面几节讨论的是电磁波在自由空间中传播的性质和它碰到界面的反透射性质.当电磁波在一定物质界面包围的空间中传播时,界面制约着其内部电磁场的存在形式,产生具有各种特殊性质的电磁波.本节和下节讨论以良导体构成的一定(至少半封闭)边界的电磁波. 我们的出发点是定态电磁波满足的波动方程,

$$(\nabla^2 + k^2)\vec{E} = 0 \qquad \nabla \cdot \vec{E} = 0 \qquad \vec{B} = -\frac{i}{\omega}\nabla \times \vec{E}$$

其中,$k^2=\omega^2\mu\epsilon$, 因为在边界内为绝缘介质. 在边界上,由于对良导体(γ很大),电磁波只能穿进导体很浅,在离表面几个穿透深度处,实际上已没有电磁场,因此,可近似认为导体内无电磁场,边条件写为,

$$\vec{n}\times\vec{E}=0 \qquad \vec{n}\times\vec{H}=\vec{i}_c \qquad \vec{n}\cdot\vec{D}=\sigma_f \qquad \vec{n}\cdot\vec{B}=0$$

它反应在导体表面上电场强度垂直于表面,磁感应强度与表面相切.

- 矩形谐振腔

  将坐标原点选在矩形谐振腔的一个角,以相邻的三个边为坐标轴. 设$u(x,y,z)$为E_x,E_y,E_z中的任意一个,则,$\nabla^2u+k^2u=0$,由分离变量法, $u(x,y,z)=X(x)Y(y)Z(z)$,

$$\nabla^2u+k^2u=YZ\frac{\partial^2X}{\partial x^2}+XZ\frac{\partial^2Y}{\partial y^2}+XY\frac{\partial^2Z}{\partial z^2}+(k_x^2+k_y^2+k_z^2)XYZ=0$$

  其中$k^2=k_x^2+k_y^2+k_z^2$,要求

$$\frac{d^2X}{dx^2}+k_x^2X=0 \qquad \frac{d^2Y}{dy^2}+k_y^2Y=0 \qquad \frac{d^2Z}{dz^2}+k_z^2Z=0$$
$$\to u(x,y,z)=(C_1\cos k_xx+D_1\sin k_xx)(C_2\cos k_yy+D_2\sin k_yy)(C_3\cos k_zz+D_3\sin k_zz)$$

  由$x=0,y=0,z=0$处边界条件得

$$E_x=A_1\cos k_xx\sin k_yy\sin k_zz$$
$$E_y=A_2\sin k_xx\cos k_yy\sin k_zz$$
$$E_z=A_3\sin k_xx\sin k_yy\cos k_zz$$

  再由$x=L_1,y=L_2,z=L_3$(L_1,L_2,L_3为矩形谐振腔的边长)处的边条件得

$$k_x=\frac{m\pi}{L_1} \qquad k_y=\frac{n\pi}{L_2} \qquad k_z=\frac{p\pi}{L_3} \qquad m,n,p=0,1,2,\cdots$$
$$\omega^2\mu\epsilon=k^2=k_x^2+k_y^2+k_z^2=\pi^2[(\frac{m}{L_1})^2+(\frac{n}{L_2})^2+(\frac{p}{L_3})^2]$$

  方程$\nabla\cdot\vec{E}=0$给出约束:$k_xA_1+k_yA_2+k_zA_3=0$

  一组给定的(m,n,p)称为一种本征振荡,它有两个独立偏振波形(因A_1,A_2,A_3中有两个独立参数),它们的谐振频率为

$$\omega_{nmp}=\frac{\pi}{\sqrt{\mu\epsilon}}\sqrt{(\frac{n}{L_2})^2+(\frac{m}{L_2})^2+(\frac{p}{L_3})^2}$$

  波长为

$$\lambda_{nmp}=\frac{2\pi}{\omega_{mnp}\sqrt{\mu\epsilon}}=\frac{2}{\sqrt{(\frac{n}{L_2})^2+(\frac{m}{L_2})^2+(\frac{p}{L_3})^2}}$$

  它的特点是电磁场位于腔内,最低频(最大波长)的非0电磁波为(110)或(101)或(011),其波长与谐振腔线度同一量级.

  应用

  - 微波范围: 采用具有金属壁面的谐振腔产生高频振荡.
  - 光学范围: 反射镜组成光学谐振腔产生近单色的激光束.

§3.5 电磁波的定向传播

本节讨论在一维方向(z方向)可以自由传播的电磁波,在另外两维方向(x,y方向),系统由良导体构成闭合边界.具体讨论中不空和中空两种情况.

- 方程及边条件

$$\begin{cases}\nabla\times\vec{E}=i\omega\vec{B}\\ \nabla\times\vec{B}=-i\mu\epsilon\omega\vec{E}\end{cases}\qquad\begin{cases}\nabla\cdot\vec{E}=0\\ \nabla\cdot\vec{B}=0\end{cases}\qquad\vec{n}\times\vec{E}\Big|_{\text{边界}}=0$$

用下标z,t表示z方向和xy平面内的方向

$$\vec{E}=[\vec{E}_z(x,y)+\vec{E}_t(x,y)]e^{ik_zz}\qquad\nabla=\nabla_{xy}+\frac{\partial}{\partial z}\vec{e}_z$$
$$\vec{B}=[\vec{B}_z(x,y)+\vec{B}_t(x,y)]e^{ik_zz}\qquad\nabla_{xy}=\frac{\partial}{\partial x}\vec{e}_x+\frac{\partial}{\partial y}\vec{e}_y$$

则场方程化为

$$\begin{cases}-i\mu\epsilon\omega\vec{E}_z=\nabla_{xy}\times\vec{B}_t\\ i\omega\vec{B}_z=\nabla_{xy}\times\vec{E}_t\end{cases}\qquad\begin{cases}-i\mu\epsilon\omega\vec{E}_t=\nabla_{xy}\times\vec{B}_z+ik_z\vec{e}_z\times\vec{B}_t\\ i\omega\vec{B}_t=\nabla_{xy}\times\vec{E}_z+ik_z\vec{e}\times\vec{E}_t\end{cases}\qquad\begin{cases}\nabla_{xy}\cdot\vec{E}_t+ik_zE_z=0\\ \nabla_{xy}\cdot\vec{B}_t+ik_zB_z=0\end{cases}$$

  - 边界(管壁)上的电流

$$\vec{i}_c=\vec{n}\times\vec{H}=\frac{1}{\mu}\vec{n}\times\vec{B}=\frac{1}{\mu}\vec{n}\times(\vec{B}_t+\vec{B}_z)e^{i(k_zz-\omega t)}$$

    注意边界上电流与磁力线正交
  - 边界(管壁)上的电荷

$$\sigma_f=\vec{n}\cdot\vec{D}=\epsilon\vec{n}\cdot\vec{E}=\epsilon\vec{n}\cdot\vec{e}_te^{i(k_zz-\omega t)}$$

- 解

以上方程组的任何一组解都可化为下列三种解的线性迭加

1. $E_z=B_z=0$
2. $E_z=0$
3. $B_z=0$

我们以下分别讨论它们

1. $E_z=B_z=0$ TEM波

$$\begin{cases}\nabla_{xy}\cdot\vec{E}_t=0\\ \nabla_{xy}\times\vec{E}_t=0\end{cases}\qquad\begin{cases}\nabla_{xy}\cdot\vec{B}_t=0\\ \nabla_{xy}\times\vec{B}_t=0\end{cases}\qquad\begin{cases}\vec{E}_t=-\frac{k_z}{\mu\epsilon\omega}\vec{e}_z\times\vec{B}_t\\ \frac{\omega}{k_z}\vec{B}_t=\vec{e}_z\times\vec{E}_t\end{cases}\qquad\vec{n}\times\vec{E}_t\Big|_{\text{边界}}=0$$

   它进一步化为

$$\vec{E}_t=-\nabla_{xy}\phi_t\qquad\nabla^2_{xy}\phi_t=0\qquad\phi_t\Big|_{\text{边界}}=\text{常数}$$

   求解
   - 若中空: 由唯一性定理ϕ_t =常数恒是解,即:$\vec{E}_t=\vec{B}_t=0$.

– 若中不空
  * 若内外边界电势相同,则ϕ_t =常数仍恒是解.
  * 若内外边界电势不同,则问题化为一个二维静电问题:在z一定的横截面内,$\vec{E}$是一个两维势场,ϕ_t在边界上等势

$$\vec{i}_c = \frac{1}{\mu}\vec{n}\times(\vec{B}_t+\vec{B}_z)e^{i(k_z z-\omega t)} = \frac{1}{\mu}\vec{n}\times(\frac{k_z}{\omega}\vec{e}_z\times\vec{E}_t)e^{i(k_z z-\omega t)} = \frac{k_z}{\mu\epsilon}\vec{e}_z\vec{n}\cdot\vec{E}_t e^{i(k_z z-\omega t)}$$
$$= -\frac{k_z}{\mu\omega}\vec{e}_z\frac{\partial\phi_t}{\partial n}e^{i(k_z z-\omega t)}$$
$$\sigma_f = \epsilon\vec{n}\cdot\vec{E}_t e^{i(k_z z-\omega t)} = \epsilon\frac{\partial\phi_t}{\partial n}e^{i(k_z z-\omega t)}$$
$$\vec{i}_c = -\frac{k_z}{\mu\epsilon\omega}\sigma_f\vec{e}_z$$

2. $E_z = 0$　　TE波

$$\begin{cases}\nabla_{xy}\cdot\vec{E}_t = 0\\ \nabla_{xy}\times\vec{E}_t = i\omega\vec{B}_z\end{cases}\quad\begin{cases}\nabla_{xy}\cdot\vec{B}_t = -ik_z B_z\\ \nabla_{xy}\times\vec{B}_t = 0\end{cases}\quad\begin{cases}\frac{\omega}{k_z}\vec{B}_t = \vec{e}_z\times\vec{E}_t\\ \vec{E}_t = -\frac{k_z}{\mu\epsilon\omega}\vec{e}_z\times\vec{B}_t + \frac{i}{\mu\epsilon\omega}\nabla_{xy}\times\vec{B}_z\end{cases}\quad \vec{n}\times\vec{E}_t\Big|_{\text{边界}} = 0$$

它进一步化为

$$\frac{\omega}{k_z}\vec{B}_t = \vec{e}_t\times(-\frac{k_z}{\mu\epsilon\omega}\vec{e}_z\times\vec{B}_t + \frac{i}{\mu\epsilon\omega}\nabla_{xy}\times\vec{B}_z) = \frac{k_z}{\mu\epsilon\omega}\vec{B}_t + \frac{i}{\mu\epsilon\omega}\nabla_{xy}B_z$$
$$\vec{B}_t = \frac{i}{\mu\epsilon\omega}\frac{1}{\frac{\omega}{k_z}-\frac{k_z}{\mu\epsilon\omega}}\nabla_{xy}B_z = \frac{ik_z}{\mu\epsilon\omega^2-k_z^2}\nabla_{xy}B_z$$
$$\vec{E}_t = -\frac{k_z}{\mu\epsilon\omega}\frac{ik_z}{\mu\epsilon\omega^2-k_z^2}\vec{e}_z\times\nabla_{xy}B_z + \frac{i}{\mu\epsilon\omega}\nabla_{xy}\times\vec{B}_z = \frac{-i}{\mu\epsilon\omega}(\frac{k_z^2}{\mu\epsilon\omega^2-k_z^2}+1)\vec{e}_z\times\nabla_{xy}B_z$$
$$= \frac{-i\omega}{\mu\epsilon\omega^2-k_z^2}\vec{e}_z\times\nabla_{xy}B_z$$
$$\frac{ik_z}{\mu\epsilon\omega^2-k_z^2}\nabla_{xy}^2B_z = -ik_zB_z \quad\rightarrow\quad (\nabla_{xy}^2+\mu\epsilon\omega^2-k_z^2)B_z = 0$$
$$\vec{n}\times\vec{E}_t\Big|_{\text{边界}} = \frac{-i\omega}{\mu\epsilon\omega^2-k_z^2}\vec{n}\cdot\nabla_{xy}B_z\Big|_{\text{边界}} = 0 \quad\rightarrow\quad \frac{\partial B_z}{\partial n}\Big|_{\text{边界}} = 0$$

注意,k_z必须为实才对应沿z方向的传播解. 这要求

$$k_z = \sqrt{\mu\epsilon\omega^2-k_x^2-k_y^2}\geq 0 \quad\rightarrow\quad \omega\geq\frac{k_x^2+k_y^2}{\mu\epsilon}\qquad\text{截止频率}$$

最后

$$\vec{i}_c = \frac{1}{\mu}\vec{n}\times(\vec{B}_t+\vec{B}_z)e^{i(k_z z-\omega t)} = \frac{1}{\mu}(\frac{ik_z}{\mu\epsilon\omega^2-k_z^2}\vec{n}\times\nabla_{xy}+\vec{e}_z)E_z e^{i(k_z z-\omega t)}$$
$$\sigma_f = \epsilon\vec{n}\cdot\vec{E}_t e^{i(k_z z-\omega t)} = \frac{-i\omega\epsilon}{\mu\epsilon\omega^2-k_z^2}(\vec{n}\times\vec{e}_z)\cdot\nabla_{xy}B_z e^{i(k_z z-\omega t)}$$

3. $B_z = 0$　　TM波(见作业).

- 矩形波导

  设矩形两边长分别为a, b, 将坐标原点取在矩形的一个角,两相邻的边分别为xy轴,

  1. 因为中空,无TEM波
  2. 对TE波,

$$(\nabla_{xy}^2 + \mu\epsilon\omega^2 - k_z^2)B_z = 0 \quad \rightarrow \quad B_z = (a_1 \sin k_x x + b_1 \cos k_x x)(a_2 \sin k_y y + b_2 \cos k_y y)$$
$$\frac{\partial B_z}{\partial n}\Big|_{\text{边界}} = 0 \quad \rightarrow \quad \frac{\partial B_z}{\partial x}\Big|_{x=0,x=a} = \frac{\partial B_z}{\partial y}\Big|_{y-0,y=b} = 0$$

  解

$$B_z = A\cos k_x x \cos k_y y \qquad k_x a = m\pi \quad k_y b = n\pi \quad m, n = 0, 1, 2, \cdots \quad (\frac{m\pi}{a})^2 + (\frac{n\pi}{b})^2 = \omega^2\mu\epsilon - k_z^2$$

  一组(m, n)叫一个模式,非零对应(m, n)最小的模式叫最低模式. 现在(1,0), (0,1)是最低模式.对给定k_z,ω随模式增高而变大.对一定的模式存在一个最低的频率

$$\omega_c = \frac{1}{\sqrt{\mu\epsilon}}\sqrt{(\frac{m\pi}{a})^2 + (\frac{n\pi}{b})^2}$$

  是临界频率,即对这个模式的任何频率$\omega \geq \omega_c$或$\lambda \leq \lambda_c = \frac{2\pi}{\sqrt{\mu\epsilon}\omega_c}$.

  3. 对TM波

$$(\nabla_{xy}^2 + \mu\epsilon\omega^2 - k_z^2)E_z = 0 \quad \rightarrow \quad E_z = (a_1 \sin k_x x + b_1 \cos k_x x)(a_2 \sin k_y y + b_2 \cos k_y y)$$
$$E_z\Big|_{\text{边界}} = 0 \quad \rightarrow \quad E_z\Big|_{x=0,x=a} = E_z\Big|_{y-0,y=b} = 0$$

  解

$$E_z = A\sin k_x x \sin k_y y \quad k_x a = m\pi \quad k_y b = n\pi \quad m, n = 0, 1, 2, \cdots \quad (\frac{m\pi}{a})^2 + (\frac{n\pi}{b})^2 = \omega^2\mu\epsilon - k_z^2$$

  现在(1,1)是最低模式.对给定k_z,ω随模式增高而变大.对一定的模式存在一个最低的频率

$$\omega_c = \frac{1}{\sqrt{\mu\epsilon}}\sqrt{(\frac{m\pi}{a})^2 + (\frac{n\pi}{b})^2}$$

  是临界频率.

  讨论

  - 同轴电缆中出现的是静态场
  - 波导管中出现的是驻波场. 有两种波形TE,TM波.